

NITSCI: Cross-Expert Lesion-Aware Attention with Ordinal Dual Heads and Cross-Dataset Transfer for Diabetic Retinopathy Grading

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Abstract

Background. Grading diabetic retinopathy (DR) from colour fundus photographs on the five-level international scale is difficult for deep networks when the target dataset is small: the cues that separate adjacent grades are a few small lesions, the grades are ordered, and the classes are imbalanced.

Methods. We present NITSCI, a grading network that couples an EfficientNetV2-S backbone with two token-level experts: a global expert (a Transformer encoder layer over all spatial tokens) and a lesion expert (the same tokens re-weighted by a learned, sparsity-regularised lesion-attention map). The experts exchange information through bidirectional cross-attention and are merged by an image-adaptive gate. A dual head predicts class probabilities and a continuous ordinal grade, which are combined into one score. The model is pretrained on APTOS-2019 ([**napptos**] images) and fine-tuned on IDRiD with five-fold cross-validation; decision thresholds are fitted on out-of-fold predictions only, and the five fold models are ensembled with flip test-time augmentation. All results are reported on a stratified held-out IDRiD test set ([**ntest**] images) that is never used for training, model selection or threshold fitting.

Results. On five-class grading NITSCI reaches an accuracy of [**acc5**]% (95% CI [**acc5ci**]) and a quadratic weighted kappa of [**qwk5**] (95% CI [**qwk5ci**]); [**within1**]% of test images are graded within one level of the reference. For referable DR (grade ≥ 2) it obtains an AUC of [**refauc**], sensitivity [**refsens**]% and specificity [**refspec**]%. Ablations quantify the contribution of APTOS pretraining, the lesion expert, cross-attention, the gate, the ordinal head and the inference-time ensemble. The model has 21.25 M parameters and needs 23.0 GFLOPs per 448×448 image.

Conclusions. *[AUTHOR: One sentence stating the main finding once results are filled, e.g. which components matter most, followed by: "External validation on independent cohorts is required before any clinical use."]*

Keywords: diabetic retinopathy; fundus image grading; lesion-aware attention; cross-attention; mixture of experts; ordinal regression; transfer learning; IDRiD

Highlights

- A global expert and a lesion expert exchange information through bidirectional cross-attention.
- An image-adaptive gate weighs global context against lesion evidence for each image.
- Classification and ordinal regression heads are fused into one continuous grade score.
- Thresholds are fitted on out-of-fold data; the test set is never used for any choice.
- Bootstrap confidence intervals and paired McNemar tests accompany every comparison.

1 Introduction

Diabetic retinopathy (DR) is a microvascular complication of diabetes mellitus and one of the leading causes of preventable vision loss in working-age adults. Pooled population studies estimate that roughly one in three people with diabetes has some degree of DR and about one in ten has a vision-threatening form [1]; a more recent meta-analysis puts the global prevalence among people with diabetes at about 22% and projects that the absolute number of affected people will keep rising through 2045 [2]. Because early DR is asymptomatic and timely laser or anti-VEGF treatment prevents most severe vision loss, international guidelines recommend regular screening of every person with diabetes [3].

Screening is usually performed with colour fundus photography. A trained grader assigns each image to one of the five levels of the International Clinical Diabetic Retinopathy (ICDR) severity scale [4]: no apparent retinopathy (grade 0), mild non-proliferative DR with microaneurysms only (grade 1), moderate non-proliferative DR (grade 2), severe non-proliferative DR (grade 3) and proliferative DR (grade 4). Grades 2 and above are generally considered *referable* and require specialist review. Manual grading is accurate when done by experienced readers, but it is slow, requires a scarce workforce, and shows non-negligible inter-grader variability [5]. In many low- and middle-income regions, where the diabetic population is growing fastest, the number of trained graders is far below what population screening would require [3].

Deep learning has changed what is technically possible. Convolutional neural networks trained on tens of thousands of graded photographs reach sensitivity and specificity comparable to ophthalmologists for referable DR [6–8], systems trained on very large cohorts cover the whole disease spectrum [9], and an autonomous AI system has been evaluated in a prospective pivotal trial in primary care [10]. A systematic review found the diagnostic performance of deep learning to be broadly equivalent to that of health-care professionals, while warning that few studies used external validation or reported their methods completely [11].

Why grading on small datasets remains hard

These successes rely on large, often private datasets. Many research groups and most new clinical sites have access only to a few hundred labelled images, such as the Indian Diabetic Retinopathy Image Dataset (IDRiD) [12, 13]. In this regime three properties of the task become limiting.

(i) Small, sparse evidence. Adjacent ICDR grades are separated by small findings. A single microaneurysm, a few pixels wide in a downsampled image, moves an image from grade 0 to grade 1; the number and distribution of haemorrhages and venous beading separate grade 2 from grade 3; and neovascularisation defines grade 4. A network that summarises the whole image by global average pooling gives every location the same weight, so a handful of lesion pixels can be diluted by the large area of healthy retina [14, 15].

(ii) Ordered labels. The grades are ordinal. Predicting grade 4 for a healthy eye is clinically far worse than predicting grade 1, but standard cross-entropy penalises both errors equally. The quadratic weighted kappa (QWK) used by most benchmarks [16] reflects this ordering, and so should the model and its decision rule [17–19].

(iii) Imbalance and instability. Mild and severe grades are under-represented, and with few images per class both the learned weights and any post-hoc decision thresholds are noisy. Small datasets also make it tempting, and easy, to tune choices on the test set, which inflates reported performance [20, 21].

Our approach

We address these three issues jointly in a compact model, NITSCI, and in an evaluation protocol designed for small datasets. For (i), the backbone features are processed by two experts: a *global expert* that relates all retinal regions through self-attention, and a *lesion expert* that sees the same features re-weighted by a learned lesion-attention map. Rather than fusing them by

simple concatenation, the experts exchange information through bidirectional cross-attention, so that lesion evidence is interpreted in its anatomical context and global context is focused on the lesions. An image-adaptive gate, in the spirit of mixtures of experts [22, 23], decides how much each expert contributes to the final representation. For (ii), a classification head and an ordinal regression head are trained together and combined into one continuous grade score. For (iii), the model is first trained on the larger APTOS-2019 dataset [24] and then fine-tuned on IDRiD with five-fold cross-validation; class-balanced loss weights, weight averaging and a fold ensemble with test-time augmentation stabilise the predictions; and every data-dependent choice, including the decision thresholds, is made on out-of-fold predictions while a held-out test set is used exactly once.

Contributions. The main contributions of this work are:

1. **A cross-expert lesion-aware head.** We propose a head that splits backbone tokens into a global and a lesion stream, couples them through bidirectional cross-attention, and fuses them with an image-adaptive gate. The lesion stream is learned from grade labels alone, without lesion annotations. The head adds 1.40 M parameters to a 19.85 M-parameter EfficientNetV2-S backbone.
2. **An ordinal dual-head decision rule.** The expected grade of the classification head and the output of a regression head are averaged into a continuous score whose four cut-points are fitted on out-of-fold data.
3. **A leakage-free protocol for small datasets.** Cross-dataset pretraining, stratified five-fold fine-tuning, an ensemble with test-time augmentation, and a held-out test set that is never used for training, model selection or threshold fitting; results are reported with bootstrap confidence intervals and paired McNemar tests.
4. **Component-level evidence.** Seven controlled ablations, run with identical splits and seeds, isolate the effect of the lesion map, cross-attention, gate, ordinal head, pretraining and pre-processing on both five-class grading and referable-DR screening.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the datasets, data partitioning and pre-processing. Section 4 presents the NITSCI architecture, training objective and decision rule. Section 5 gives implementation details, metrics and statistical methods. Section 6 reports the results and ablations, Section 7 discusses their implications and limitations, and Section 8 concludes.

2 Related work

This section reviews four lines of work that NITSCI builds on: deep learning for DR grading, lesion-aware and attention-based models, ordinal and imbalance-aware learning, and training strategies for small medical datasets. Comprehensive surveys of deep learning for DR are available in Tsiknakis et al. [25].

2.1 Deep learning for DR detection and grading

Early applications of convolutional neural networks (CNNs) to fundus photographs showed that end-to-end learning could replace hand-crafted lesion detectors [26, 27]. The study of Gulshan et al. [6] trained an Inception-style network on more than 100 000 graded images and reported specialist-level performance for referable DR, and similar results were obtained on independent and multi-ethnic cohorts [7, 8]. Later systems extended the task to the full disease spectrum and to lesion-level outputs [9]. In parallel, general-purpose backbones improved steadily: residual networks [28], compound-scaled EfficientNets [29], the faster-training EfficientNetV2 family [30], modernised CNNs [31], and Vision Transformers [32, 33]. Most recently, retinal foundation models pretrained by self-supervision on hundreds of thousands of unlabelled images have improved data efficiency across retinal tasks [34].

Public benchmarks have made methods comparable. Messidor [35], DDR [36], APTOS-2019 [24] and IDRiD [12] differ in population, camera, image quality and labelling protocol. IDRiD is particularly valuable because it combines image-level grades with pixel-level lesion annotations for a subset of images, and its grading challenge established an official partition and leaderboard [13]. Its small size, however, makes it a demanding test of data efficiency. Uncertainty-aware grading [37] and studies of reference-standard quality [5] have further shown that reported accuracy depends strongly on how labels and test sets are constructed.

2.2 Lesion-aware and attention-based grading

Because DR grades are defined by lesions, many methods try to make the network focus on them. Zoom-in-Net [14] produces attention maps that highlight suspicious regions and re-examines them at higher resolution. Deep image mining [38] derives lesion heat-maps from a grading network trained with image-level labels only. Collaborative learning [39] trains lesion segmentation and grading jointly, using a small set of lesion masks to guide the grading network. CANet [40] uses attention to model the relation between DR and diabetic macular oedema, and CABNet [41] introduces a category attention block that learns discriminative regions for each grade under class imbalance. Lesion-aware transformers [15] learn lesion queries with a transformer decoder and use them to predict the grade.

Channel and spatial attention modules such as squeeze-and-excitation [42] and CBAM [43], as well as non-local blocks [44], are generic ways to re-weight features. Self-attention in Transformers [45] relates all positions at once, and cross-attention between two token streams has been used to fuse multi-scale branches in CrossViT [46]. Dedicated lesion segmentation networks, from U-Net [47] to recent heterogeneous Mamba-based ensembles [48, 49], produce explicit lesion maps but require pixel-level annotations that are expensive to obtain.

NITSCI sits between these families. Like deep image mining it learns a lesion-attention map from grade labels alone; like CrossViT it uses cross-attention between two token streams; and like mixture-of-experts models [22, 23] it combines the streams with a learned gate. The difference from single-map attention modules is that the lesion map defines a *second expert* whose tokens interact with the global expert in both directions, rather than only re-weighting a single pooled feature.

2.3 Ordinal and imbalance-aware learning

Ordinal classification can be reduced to a series of binary “greater than k ” problems [50], an idea later implemented with multiple CNN outputs [17] and made rank-consistent by CORAL [18]. Other work optimises the evaluation metric directly, for example with a differentiable weighted-kappa loss [19]. In DR competitions a simple and robust alternative has been to regress the grade as a real number and to optimise the cut-points between grades on validation data; NITSCI follows this practice but combines the regression output with the expected value of the classification head so that both training signals contribute to the decision.

Class imbalance is usually handled by re-weighting or re-sampling. Focal loss [51] down-weights easy examples, and class-balanced loss [52] re-weights classes by their effective number of samples. Label smoothing [53] reduces over-confidence, which is useful when a few labels may be noisy. We use inverse-square-root class weights with label smoothing, a moderate choice that does not over-amplify the rarest grades.

2.4 Learning from small medical datasets

Transfer learning from ImageNet [54] is the default initialisation for medical image classification. Better ImageNet models tend to transfer better [55], and larger pretraining corpora such as ImageNet-21k help further [56], but the benefit for medical images depends on the domain gap and model size [57]. An intermediate fine-tuning stage on a larger in-domain dataset is a natural way to narrow this gap; we use APTOS-2019 for this purpose.

Several inference- and optimisation-level techniques improve robustness at little cost. Ensembles of independently trained networks improve accuracy and uncertainty estimates [58]; averaging weights along the training trajectory, through exponential moving averages [59] or stochastic weight averaging [60], finds flatter solutions that generalise better; and test-time augmentation (TTA) reduces prediction variance in medical imaging [61]. Modern networks are often poorly calibrated [62], which matters when a probability threshold defines a referral decision.

Finally, evaluation practice is critical in small-data settings. Selecting models or thresholds with the same data used to estimate performance biases the estimate upward [20], and data leakage is a widespread source of irreproducible results in machine-learning-based science [21]. Reporting guidelines such as TRIPOD+AI [63] ask for a clear separation of development and evaluation data, confidence intervals and transparent reporting of all modelling choices. Our protocol is designed to meet these requirements.

3 Materials

3.1 Datasets

IDRiD (target dataset). The Indian Diabetic Retinopathy Image Dataset [12, 13] was collected at an eye clinic in Nanded, India, with a Kowa VX-10 α fundus camera at a resolution of 4288×2848 pixels and a 50° field of view centred near the macula. Each image is graded on the five-level ICDR scale [4] by medical experts; a subset also has pixel-level lesion masks, which we do not use for training. We use the publicly distributed Kaggle copy of the grading subset. Image files are matched to label rows by file name, with a fallback that compares the digit sequence in the name; after matching and de-duplication, [nidrid] graded images remain.

APTOS-2019 (pretraining dataset). The APTOS-2019 Blindness Detection dataset [24] contains [naptos] labelled fundus photographs collected in rural India by the Aravind Eye Hospital, graded on the same 0–4 scale. Images come from several cameras and vary widely in size, illumination, focus and field of view, which makes the dataset a useful source of appearance diversity. It is used only for stage-1 pretraining and never for evaluation.

Table 1 summarises the grade distributions (shown graphically in Supplementary Fig. S1). Both datasets are imbalanced: grade 0 and grade 2 dominate, while grades 1 and 3 are comparatively rare. This imbalance motivates the class-weighted loss (Section 4) and the use of balanced accuracy, macro-F1 and QWK alongside plain accuracy (Section 5).

Table 1: Grade distribution of the datasets used. The IDRiD test split is a stratified 15% hold-out that is never used for training, model selection or threshold fitting.

Dataset	Grade 0	Grade 1	Grade 2	Grade 3	Grade 4	Total
APTOS-2019 (pretraining)	[nap0]	[nap1]	[nap2]	[nap3]	[nap4]	[naptos]
IDRiD (all)	[nid0]	[nid1]	[nid2]	[nid3]	[nid4]	[nidrid]
IDRiD test (held out)	[nte0]	[nte1]	[nte2]	[nte3]	[nte4]	[ntest]

3.2 Data partitioning

Figure 1 summarises the protocol. Before any model is trained, a stratified 15% of IDRiD ([ntest] images, random seed 42) is set aside as the test set. It is used exactly once, after all modelling decisions have been fixed: it plays no role in training, early stopping, checkpoint selection, threshold fitting, choice of the decision rule or any hyper-parameter. The remaining [ntrainval] images are divided into five stratified folds. In each fold, four parts are used for training and one for validation (early stopping and checkpoint selection); the validation predictions of all five folds together form the out-of-fold (OOF) set on which the decision rule and thresholds are fitted. For APTOS-2019, a separate stratified 90/10 split is used for checkpoint selection during pretraining.

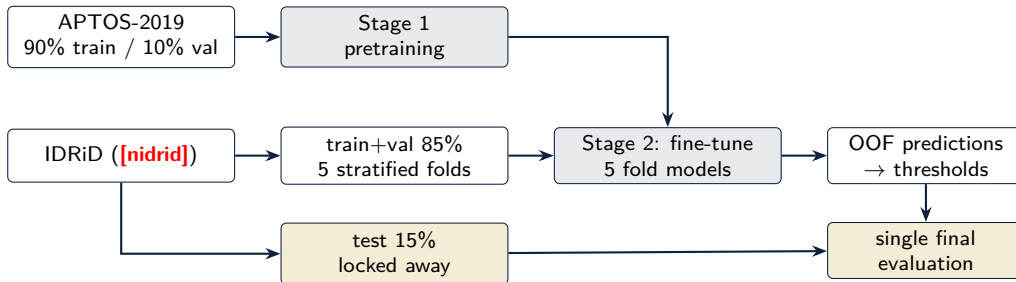


Figure 1: Data protocol. The IDRiD test split (gold) is separated before training and enters only the final evaluation; all data-dependent decisions use APTOS or the out-of-fold IDRiD predictions.

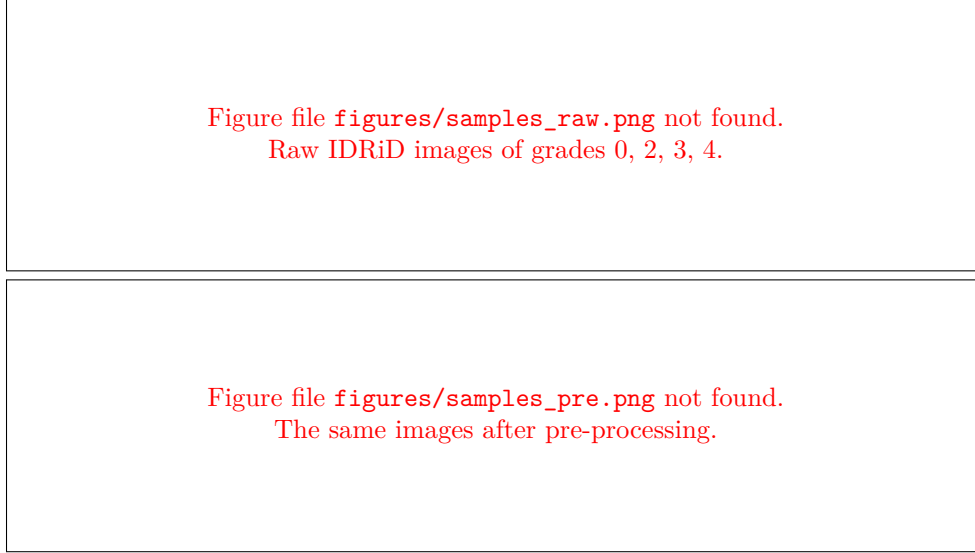


Figure 2: Example IDRiD images before (top) and after (bottom) cropping, padding and Ben Graham normalisation.

3.3 Pre-processing

Fundus photographs from different cameras differ in resolution, field of view, black borders and illumination. Each image is processed identically for both datasets (Fig. 2):

1. images whose longer side exceeds 1600 pixels are first down-scaled with area interpolation, which keeps the remaining steps fast without affecting the final 448-pixel resolution;
2. the black border is removed by cropping to the bounding box of pixels whose grey level exceeds 10;
3. the image is zero-padded to a square so that the circular fundus is not distorted, and resized to 448×448 pixels;
4. Ben Graham’s local-contrast normalisation [64] is applied,

$$I_{\text{ben}} = 4I - 4(G_{\sigma} * I) + 128, \quad \sigma = 448/30, \quad (1)$$

where $G_{\sigma} *$ denotes Gaussian blurring; subtracting the local mean removes slow illumination gradients and camera-specific colour casts, while the factor 4 amplifies local contrast such as small red lesions and exudates;

5. pixels outside a centred circle of radius 0.48×448 are set to grey (128) to suppress the bright ring artefact that Eq. (1) creates at the edge of the fundus.

As an alternative pre-processing, evaluated in the ablation study, we apply contrast-limited adaptive histogram equalisation (CLAHE) [65] to the green channel (clip limit 2, 8×8 tiles), where vessels and red lesions have the highest contrast, followed by an edge-preserving bilateral filter. Pre-processed images are cached once and then normalised with ImageNet channel statistics.

3.4 Data augmentation

To reduce over-fitting on a few hundred images, training images are augmented on the fly with random horizontal and vertical flips; random rotation in $[-180^{\circ}, 180^{\circ}]$, which is valid because the circular fundus has no canonical orientation once the image is padded; translation of up to 3%; scaling by a factor in $[0.95, 1.10]$; colour jitter (brightness and contrast 0.2, saturation 0.1, hue 0.02); and random erasing [66] of 2–8% of the image with probability 0.2, which discourages reliance on a single region. Validation and test images are not augmented, except for the deterministic flips used for test-time augmentation (Section 4).

4 Method

This section describes the NITSCI model (Fig. 3), its training objective, the two-stage training procedure and the decision rule. Given an image $x \in \mathbb{R}^{3 \times 448 \times 448}$, a backbone produces a feature map that is turned into N tokens. Two experts process the tokens, exchange information, and are pooled and fused by a gate; two heads produce a class distribution and a continuous grade.

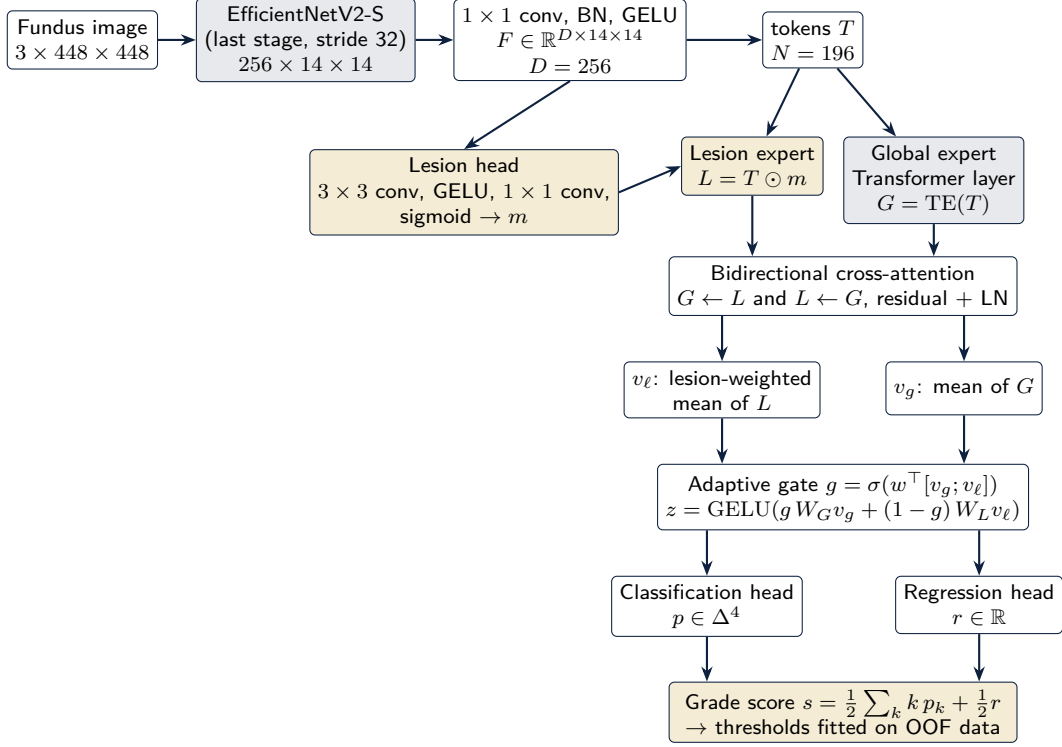


Figure 3: NITSCI architecture. Blue: backbone and global expert. Gold: lesion pathway and decision score. Token counts are for a 448×448 input. The lesion map m is learned from grade labels only.

4.1 Backbone and tokenisation

We use EfficientNetV2-S [30] pretrained on ImageNet-21k and fine-tuned on ImageNet-1k [56], taken from the `timm` library [67]. Its last stage (stride 32, 256 channels) gives a 14×14 map for a 448×448 input. A 1×1 convolution with batch normalisation and GELU projects it to $F \in \mathbb{R}^{D \times h \times w}$ with $D = 256$, which is flattened into $N = hw = 196$ tokens $T \in \mathbb{R}^{N \times D}$.

4.2 Lesion-attention map and lesion expert

A light head (3×3 convolution to $D/4$ channels, GELU, 1×1 convolution to one channel, sigmoid) predicts a spatial map $m \in [0, 1]^{h \times w}$. No lesion annotation is used: m is learned end-to-end from grade labels and is encouraged to be sparse by an ℓ_1 penalty (Eq. 7). The lesion expert is the re-weighted token set

$$L = T \odot m, \quad L_i = m_i T_i, \quad i = 1, \dots, N. \quad (2)$$

4.3 Global expert

The global expert is one pre-norm Transformer encoder layer [45] with four heads, a feed-forward width of $2D$ and dropout 0.1, applied to all tokens: $G = \text{TE}(T)$. It models context across the whole retina, such as the spatial spread of lesions and their relation to the optic disc and macula.

4.4 Bidirectional cross-attention

Each expert queries the other with multi-head attention (four heads, dropout 0.1):

$$Z_g = \text{MHA}(Q=G, K=L, V=L), \quad G' = \text{LN}(G + Z_g), \quad (3)$$

$$Z_\ell = \text{MHA}(Q=L, K=G, V=G), \quad L' = \text{LN}(L + Z_\ell). \quad (4)$$

The global stream thus gathers lesion evidence, and the lesion stream receives the context needed to interpret it.

4.5 Pooling and adaptive gate

The global stream is mean-pooled, $v_g = \frac{1}{N} \sum_i G'_i$, and the lesion stream is pooled with the lesion weights, $v_\ell = \sum_i m_i L'_i / (\sum_i m_i + \epsilon)$. A scalar gate decides, per image, how much each expert contributes:

$$g = \sigma(w^\top [v_g; v_\ell] + b), \quad z = \text{GELU}(g W_G v_g + (1 - g) W_L v_\ell). \quad (5)$$

This is a two-expert soft mixture [22]; g is also reported for interpretability.

4.6 Dual head and grade score

Two heads share z : a classification head (LayerNorm, dropout 0.3, linear) gives logits and probabilities $p \in \Delta^4$, and a regression head of the same form gives $r = f_r(z) + 2$, centred on the middle grade. Their outputs are combined into a continuous score

$$s = \frac{1}{2} \sum_{k=0}^4 k p_k + \frac{1}{2} r. \quad (6)$$

A grade is obtained by comparing s with four ordered thresholds $\theta_1 < \dots < \theta_4$: $\hat{y} = \#\{j : s \geq \theta_j\}$.

4.7 Training objective

For a mini-batch the loss is

$$\mathcal{L} = \text{CE}_{w,\varepsilon}(p, y) + \lambda_r \text{SmoothL1}(r, y) + \lambda_m \bar{m}, \quad (7)$$

where $\text{CE}_{w,\varepsilon}$ is cross-entropy with label smoothing $\varepsilon = 0.05$ [53] and class weights $w_c \propto n_c^{-1/2}$ (normalised to mean one, computed on the training fold), $\bar{m}$ is the mean of the up-sampled lesion map, $\lambda_r = 0.5$ and $\lambda_m = 0.01$. The square-root weighting is a compromise between ignoring and fully inverting the imbalance.

4.8 Two-stage training

Stage 1 (APTOS). The full model, initialised from ImageNet-21k backbone weights, is trained on APTOS-2019 for 10 epochs (learning rate 3×10^{-4} , EMA decay 0.999). **Stage 2 (IDRiD).** For each of the five folds, a model initialised from the stage-1 weights is fine-tuned for at most 25 epochs (learning rate 2×10^{-4} , EMA decay 0.99, early stopping with patience 8). In both stages we use AdamW [68] with weight decay 10^{-4} , a backbone learning rate one third of the head learning rate, a one-cycle cosine schedule with 10% warm-up [69], mixed precision [70], gradient clipping at norm 2, and an exponential moving average (EMA) of weights [59] that is used for validation and saved. The checkpoint with the highest sum of validation accuracy and quadratic weighted kappa (fixed thresholds 0.5, 1.5, 2.5, 3.5, no TTA) is kept.

4.9 Inference and decision rule

Each fold model predicts every test image under four views (identity, horizontal flip, vertical flip, both); probabilities and regression outputs are averaged over views and over the five folds (20 predictions per image). The decision rule is chosen *on out-of-fold (OOF) predictions only* among three candidates: (i) argmax of p ; (ii) rounding s with thresholds 0.5, 1.5, 2.5, 3.5; (iii) optimised thresholds, found by coordinate search (four rounds, 80 candidate values per threshold, constrained to remain ordered) maximising OOF accuracy. The chosen rule was **[rule]**, with thresholds (**[thr]**).

Referable DR. For the screening task we use $P(\text{referable}) = p_2 + p_3 + p_4$ and select its operating threshold on OOF predictions by maximising Youden’s J [71] (selected value **[refthr]**).

4.10 Model size and cost

Table 2 lists the parameter counts and floating-point operations, measured with PyTorch [72] on a single 448×448 input. The cross-expert head adds 7% to the parameters of the backbone.

Table 2: Model size and cost for one 448×448 image (FLOPs counted as $2 \times$ multiply-accumulates).

Component	Parameters (M)	Notes
EfficientNetV2-S feature extractor	19.85	256 output channels
Cross-expert head (all modules after the backbone)	1.40	$D = 256$, 4 heads
NITSCI total	21.25	23.0 GFLOPs
Plain baseline (same backbone, GAP, same dual head)	19.92	–
Inference latency per image (fp16, batch 16, [gpu])	[mspi] ms per model and view	

5 Experimental setup

5.1 Implementation details

All experiments were implemented in PyTorch [72] with the `timm` model library [67] and scikit-learn [73], and run on a single `[gpu]` GPU. Mixed precision (fp16) [70] was used for training and inference, and Grad-CAM maps were computed with the `pytorch-grad-cam` library [74]. Random seeds were fixed to 42 for data splitting and pretraining and to $42 + k$ for fold k . Table 3 lists all hyper-parameters; none of them was tuned on the test set. After early stopping, the number of epochs actually run in the five folds was (`[epochsrn]`).

Table 3: Training and inference hyper-parameters. Values marked $\dagger$ are shared by both stages.

Setting	Stage 1 (APTOS-2019)	Stage 2 (IDRiD, per fold)
Initialisation	ImageNet-21k backbone	stage-1 weights
Epochs (max.)	10	25, early stopping (patience 8)
Peak learning rate (head / backbone)	3×10^{-4} / 1×10^{-4}	2×10^{-4} / 6.7×10^{-5}
EMA decay	0.999	0.99
Optimiser †	AdamW [68], weight decay 10^{-4}	
Schedule †	one-cycle cosine, 10% warm-up [69]	
Batch size †	16; gradient clipping at ℓ_2 norm 2.0	
Input size †	448×448 , Ben Graham pre-processing	
Loss weights †	$\lambda_r = 0.5$ (ordinal), $\lambda_m = 0.01$ (sparsity), label smoothing 0.05	
Class weights †	$w_c \propto n_c^{-1/2}$, normalised to mean 1, computed on training data	
Checkpoint selection †	max. validation accuracy + QWK (EMA weights, fixed cut-points, no TTA)	
Inference	–	5 fold models $\times$ 4 flips = 20 predictions per image

5.2 Evaluation metrics

Let y_i and $\hat{y}_i$ be the reference and predicted grades of test image i , $i = 1, \dots, n$, and let O be the 5×5 confusion matrix with entries O_{jk} counting images of grade j predicted as grade k .

Five-class grading. *Accuracy* is the fraction of exactly correct grades. Because the classes are imbalanced, we also report *balanced accuracy*, the mean of the per-class recalls, and the macro-averaged precision, recall and F1, which weight every grade equally. The *quadratic weighted kappa* [16] measures agreement beyond chance while penalising errors by the squared grade distance:

$$\kappa_w = 1 - \frac{\sum_{j,k} W_{jk} O_{jk}}{\sum_{j,k} W_{jk} E_{jk}}, \quad W_{jk} = \frac{(j - k)^2}{(K - 1)^2}, \quad (8)$$

where E is the expected confusion matrix under independence (outer product of the marginal histograms, scaled to n) and $K = 5$. We also report unweighted Cohen’s kappa, the Matthews correlation coefficient (MCC), the *within-one-grade accuracy* $\frac{1}{n} \sum_i \mathbb{1}[|y_i - \hat{y}_i| \leq 1]$, which quantifies how many errors are between adjacent grades, and macro one-vs-rest ROC AUC and average precision computed from the ensemble probabilities.

Referable DR. With TP, FN, FP and TN counted for the binary task (grade ≥ 2 vs. < 2), we report sensitivity $TP/(TP + FN)$, specificity $TN/(TN + FP)$, positive and negative predictive values, accuracy, F1 and the threshold-free ROC AUC of $P(\text{referable})$. Sensitivity is the clinically most important quantity, because a missed referable case delays treatment.

5.3 Statistical analysis

Confidence intervals. For each test metric we compute a 95% percentile bootstrap interval [75]: the n test images are resampled with replacement 2000 times, the metric is recomputed on

each resample, and the 2.5th and 97.5th percentiles are reported. Predictions are fixed during resampling, so the interval reflects uncertainty due to the finite test set, not training randomness.

Paired comparisons. All configurations are evaluated on the same test images, so we compare each ablation with the full model using the two-sided exact McNemar test on per-image correctness [76]. If b images are correct only for the full model and c only for the ablation, the p -value is

$$p = \min\left(1, 2 \sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)}\right). \quad (9)$$

We report p -values without claiming significance for a single threshold; seven ablations are compared, so a Bonferroni-adjusted level of $0.05/7 \approx 0.007$ is the conservative reference.

Cross-validation estimates. We additionally report OOF metrics and their mean and standard deviation across folds. These are slightly optimistic, because each fold’s checkpoint is selected on the same fold [20]; they are given for completeness and are not used as the main result.

5.4 Compared configurations

All configurations use identical splits, seeds, augmentation, schedule and inference, and differ in one factor only:

- **Plain:** the same backbone and 1×1 projection, followed by global average pooling and the same dual head and loss. This is the standard transfer-learning baseline and isolates the effect of the whole cross-expert head.
- **w/o APTOS:** no stage-1 pretraining; the model is initialised from ImageNet-21k only.
- **CLAHE:** CLAHE + bilateral filtering instead of Ben Graham pre-processing, with stage 1 repeated.
- **w/o ordinal head:** $\lambda_r = 0$; the regression output is untrained, and the decision-rule selection on OOF data (Section 4.9) decides whether the score s or the argmax of p is used.
- **w/o lesion map:** $m \equiv 1$; the lesion expert receives all tokens with equal weight.
- **w/o cross-attention:** the experts are only normalised, $G' = \text{LN}(G)$, $L' = \text{LN}(L)$.
- **w/o gate:** the gate is fixed to $g = 0.5$, i.e. equal contribution of both experts.

For the architectural variants, stage 1 is repeated so that pretraining and fine-tuning use the same architecture.

5.5 Reproducibility

The notebook, the export cell that writes every reported number and figure, the ablation variants and the script that inserts the numbers into this manuscript are released together (see Code availability). The held-out test indices and per-image predictions are stored in the exported results, so every table can be regenerated and every comparison re-tested without retraining.

6 Results

This section reports the held-out test results for five-class grading (Section 6.1) and referable-DR screening (Section 6.2), followed by the ablation study, the effect of inference-time choices, training behaviour, visual explanations and computational cost. Unless stated otherwise, all numbers refer to the **[ntest]**-image IDRiD test set and to the five-fold ensemble with test-time augmentation.

6.1 Five-class grading

Table 4 reports the held-out test performance. Cross-validation estimates on the training portion are given for comparison: OOF accuracy **[oofacc]**% and QWK **[oofqwk]** (per-fold accuracy **[foldacc]**%, per-fold QWK **[foldqwk]**).

Table 4: Five-class DR grading on the held-out IDRiD test set ($n = \text{[ntest]}$). Ensemble of five fold models with four-view TTA, decision rule chosen on OOF data. Brackets: 95% bootstrap CI.

Metric	Value	Metric	Value
Accuracy (%)	[acc5] [acc5ci]	Macro precision (%)	[precmacro5]
Balanced accuracy (%)	[bacc5]	Macro recall (%)	[recmacro5]
Quadratic weighted kappa	[qwk5] [qwk5ci]	Macro F1 (%)	[f1macro5] [f1macro5ci]
Cohen’s kappa	[kappa5]	Weighted F1 (%)	[f1w5]
MCC	[mcc5]	Within-one-grade accuracy (%)	[within1]
Macro ROC AUC (OvR)	[auc5]	Macro average precision	[ap5]

Table 5: Per-grade precision, recall and F1 (%) on the test set.

	0 No DR	1 Mild	2 Moderate	3 Severe	4 PDR
Support	[nte0]	[nte1]	[nte2]	[nte3]	[nte4]
Precision	[p0]	[p1]	[p2]	[p3]	[p4]
Recall	[r0]	[r1]	[r2]	[r3]	[r4]
F1	[f0]	[f1]	[f2]	[f3]	[f4]

Table 5 and Fig. 4(a) show where errors occur. *[AUTHOR: Describe the confusion pattern: e.g. whether most errors are between adjacent grades (consistent with within-one-grade accuracy of **[within1]**%), and which grade is weakest. Grade 1 is typically hardest because it is defined by microaneurysms only.]*

6.2 Referable DR screening

Table 6 reports the screening task. The operating threshold was fixed on OOF data before the test set was evaluated; the OOF AUC was **[oofrefauc]**. Figure 4 (right) shows the ROC curves.

Table 6: Referable DR (grade ≥ 2 vs. < 2) on the held-out test set. Brackets: 95% bootstrap CI.

Metric	Value	Metric	Value
ROC AUC	[refauc] [refauc]	PPV (%)	[refppv]
Sensitivity (%)	[refsens] [refsensci]	NPV (%)	[refnpv]
Specificity (%)	[refspec] [refspecci]	F1 (%)	[reff1]
Accuracy (%)	[refacc] [refacc]	TP / FN / FP / TN	[tp] / [fn] / [fp] / [tn]

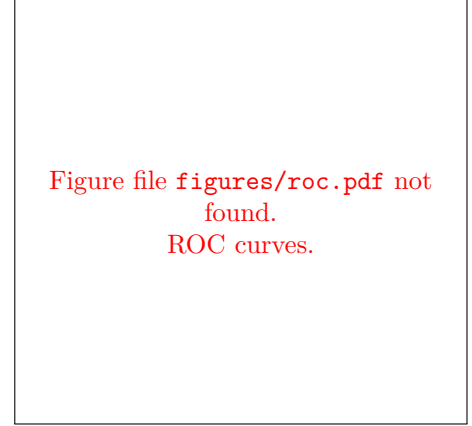
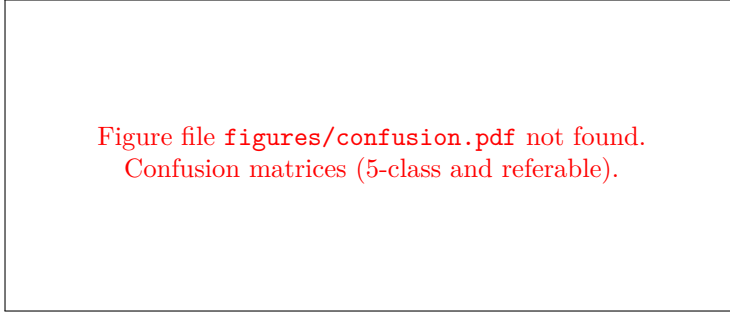


Figure 4: Held-out test set. Left: confusion matrices for (a) five-class grading and (b) referable DR (grade ≥ 2). Right: one-vs-rest ROC curves for each grade and for referable DR (bold).

6.3 Ablation study

Table 7 compares the full model with the configurations of Section 5. All runs share the same test images, so the McNemar p -value tests whether per-image correctness differs from the full model.

Table 7: Ablation study on the held-out IDRiD test set. Five-class accuracy, QWK and macro F1; referable-DR AUC, sensitivity and specificity. p : exact McNemar test vs. the full model on five-class correctness.

Configuration	Params (M)	Acc (%)	QWK	F1 (%)	Ref. AUC	Sens. (%)	Spec. (%)
NITSCI (full)	[abfullparams]	[abfullacc]	[abfullqwk]	[abfullf1]	[abfullrefauc]	[abfullrefsens]	[abfullrefspec]
Plain (GAP head)	[abplainparams]	[abplainacc]	[abplainqwk]	[abplainf1]	[abplainrefauc]	[abplainrefsens]	[abplainrefspec]
w/o APTOS pretraining	[abnoaptosparams]	[abnoaptosacc]	[abnoaptosqwk]	[abnoaptosf1]	[abnoaptosrefauc]	[abnoaptosrefsens]	[abnoaptosrefspec]
CLAHE pre-processing	[abclaheparams]	[abclaheacc]	[abclaheqwk]	[abclahef1]	[abclaherefauc]	[abclaherefsens]	[abclaherefspec]
w/o ordinal head	[abnoregparams]	[abnoregacc]	[abnoregqwk]	[abnoregf1]	[abnoregrefauc]	[abnoregrefsens]	[abnoregrefspec]
w/o lesion map	[abnolesionparams]	[abnolesionacc]	[abnolesionqwk]	[abnolesionf1]	[abnolesionrefauc]	[abnolesionrefsens]	[abnolesionrefspec]
w/o cross-attention	[abnocrossparams]	[abnocrossacc]	[abnocrossqwk]	[abnocrossf1]	[abnocrossrefauc]	[abnocrossrefsens]	[abnocrossrefspec]
w/o adaptive gate	[abnogateparams]	[abnogateacc]	[abnogateqwk]	[abnogatef1]	[abnogaterefauc]	[abnogaterefsens]	[abnogaterefspec]

McNemar p vs. full: Plain **[abplainp]**; w/o APTOS **[abnoaptosp]**; CLAHE **[abclahep]**; w/o ordinal **[abnoregp]**; w/o lesion map **[abnolesionp]**; w/o cross-attention **[abnocrossp]**; w/o gate **[abnogatep]**. Parameter counts of “w/o cross-attention” and “w/o gate” include the unused modules, which are kept so that the remaining weights are initialised identically.

[AUTHOR: Interpret Table 7 honestly: name the components whose removal lowers accuracy/QWK by more than the width of the bootstrap interval or with McNemar $p < 0.05$, and state plainly which components show no measurable effect at this test-set size. Do not claim a component “improves” performance if its p -value is large.]

6.4 Inference-time choices

Table 8 isolates the effect of the decision rule, TTA and fold ensembling without retraining. The decision rule was selected on OOF accuracy; test values are shown only to document the outcome of that choice.

6.5 Training behaviour

Supplementary Fig. S2 shows the training loss and validation metrics for each fold. *[AUTHOR: Comment on convergence, the epoch at which early stopping triggered, and the spread between folds.]*

Table 8: Effect of inference-time choices. Top: decision rules (OOF accuracy was used for selection). Bottom: single fold models vs. ensemble, with and without TTA (mean $\pm$ SD over five folds for single models).

Decision rule	OOF acc. (%)	Test acc. (%)	Test QWK
Argmax of p	[oofargmax]	[testargmax]	[testargmaxqwk]
Rounded score s	[oofrounded]	[testrounded]	[testroundedqwk]
Optimised thresholds on s	[oofopt]	[testopt]	[testoptqwk]

Prediction	Test acc. (%)	Test QWK
Single fold, TTA	[singleacc]	[singleqwk]
5-fold ensemble, no TTA	[notaacc]	[notaqwk]
5-fold ensemble + TTA	[ttaacc]	[ttaqwk]

6.6 Visual explanations

Figure 5 shows Grad-CAM maps for the predicted grade and the learned lesion-attention map m for test images, together with the gate value g (Eq. 5). Because m is learned without lesion annotations, these maps are qualitative only; we do not claim that they localise lesions. *[AUTHOR: Describe what the maps show, e.g. whether high attention falls on haemorrhages or exudates, and whether g differs between low and high grades. A quantitative check against the IDRiD lesion masks (e.g. pointing-game accuracy) would strengthen this section.]*

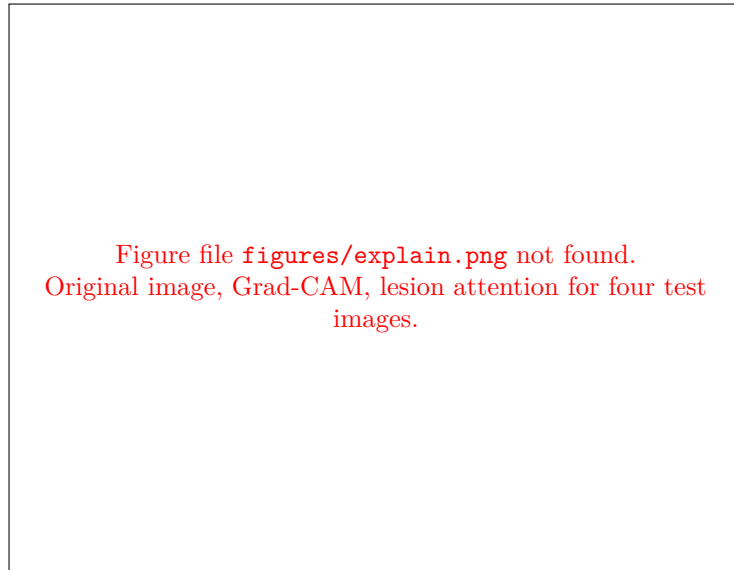


Figure 5: Test images (top), Grad-CAM for the predicted grade (middle), and the learned lesion-attention map with gate value g (bottom), from the fold-1 model.

6.7 Computational cost

The full model has 21.25 M parameters and requires 23.0 GFLOPs per 448×448 image (Table 2). Measured inference time on the [gpu] was [mspi] ms per image per model and view, so the 20-prediction ensemble costs about twenty times this value per image. *[AUTHOR: State whether this is acceptable for screening throughput, and note that a single fold model without TTA is the cheaper option whose accuracy is given in Table 8.]*

7 Discussion

7.1 Main findings

On a held-out IDRiD test set that played no role in any modelling decision, NITSCI reached a five-class accuracy of **[acc5]**% with a QWK of **[qwk5]**, graded **[within1]**% of images within one level of the reference, and detected referable DR with an AUC of **[refauc]** at a sensitivity of **[refsens]**% and a specificity of **[refspec]**%. *[AUTHOR: Summarise in two or three sentences, using only claims supported by Tables 4–8: (1) whether the full model improves on the plain baseline beyond the bootstrap interval; (2) which ablations had the largest effect; (3) whether the ordinal head and threshold fitting reduced large errors, visible in QWK and within-one-grade accuracy.]*

7.2 Interpretation of the components

Cross-dataset pretraining. APTOS-2019 differs from IDRiD in cameras, image quality and population, but shares the grading scale and the visual vocabulary of DR lesions. Pretraining on it exposes the model to roughly eight times more graded fundus images before it sees IDRiD, which is consistent with the general observation that in-domain data narrows the gap left by ImageNet pretraining [57]. *[AUTHOR: Report the size of the “w/o APTOS” effect from Table 7 and whether it is the largest single factor.]*

Lesion expert, cross-attention and gate. The three architectural components address the same problem from different angles: the lesion map concentrates the representation on a few informative locations, cross-attention lets lesion tokens be interpreted in their retinal context, and the gate allows the balance between global and lesion evidence to vary from image to image. A plausible expectation is that the gate favours lesion evidence for borderline images between grades 0, 1 and 2, where the decision depends on a few lesions, and global context for advanced disease, where lesions are widespread. *[AUTHOR: Check this with the gate values exported for the test images, or remove the sentence. Report which of the three ablations changed accuracy or QWK beyond the confidence interval, and say plainly if a component had no measurable effect at this test-set size.]*

Ordinal head and decision rule. Combining the expected grade of the classifier with a regression output produces a score that changes smoothly with disease severity, so errors tend to fall on adjacent grades. Fitting the cut-points on out-of-fold predictions adapts them to the class balance of IDRiD without touching the test set. The comparison of decision rules in Table 8 shows how much of the final accuracy is due to the rule rather than to the network. *[AUTHOR: Comment on whether the OOF ranking of the rules matched their test ranking; if not, this indicates that the threshold fit partly reflects noise in a small OOF set.]*

Ensembling and TTA. Averaging 20 predictions per image reduces variance, which matters most for the under-represented grades. The difference between a single fold model and the ensemble (Table 8) also quantifies a practical trade-off: a single model without TTA is twenty times cheaper at inference.

7.3 Comparison with published results

Direct numerical comparison with the IDRiD grading challenge [13] is not possible from this study, because the challenge uses an official train/test partition whereas we use a stratified random hold-out of the Kaggle copy. Numbers reported on different splits, with different pre-processing and sometimes with thresholds tuned on the test set, are not comparable, and placing them in the same table would mislead the reader. We therefore compare against a plain baseline and ablations

trained under identical conditions (Table 7). *[AUTHOR: Either (a) re-run the pipeline on the official IDRiD grading partition and add a table comparing with the challenge results reported in Porwal et al. [13], copying the numbers from that paper; or (b) keep this paragraph.]*

7.4 Interpretability

The lesion-attention map and Grad-CAM [77] offer a view of where the model looks, which can help clinicians to spot obvious failure modes. They should not be over-interpreted: saliency maps can be insensitive to the model parameters [78], and post-hoc explanations of black-box models are not a substitute for validated, inherently interpretable evidence in high-stakes decisions [79]. Because our lesion map is learned without lesion masks, we treat it as a qualitative aid only; validating it against the IDRiD lesion annotations is part of future work.

7.5 Limitations

- **Small test set.** With **[ntest]** test images, confidence intervals are wide, and per-grade metrics for rare grades rest on few images. Differences of a few percentage points between configurations may not be statistically detectable.
- **Single target dataset.** All results come from IDRiD, collected at one centre with one camera. Performance on other populations, cameras and image-quality levels, for example Messidor [35] or DDR [36], has not been tested.
- **Non-official split.** The random hold-out prevents direct comparison with the IDRiD challenge.
- **Weakly supervised lesion map.** The lesion-attention map has not been validated against lesion masks.
- **Training randomness.** Variance is reflected through five folds but not through repeated runs with different seeds.
- **Reference standard.** Grades are taken from the dataset providers; grader variability, which is substantial for DR [5], is not modelled.
- **Optimistic OOF estimates.** Checkpoints are selected on the same folds used for OOF predictions, so the cross-validation numbers (but not the test numbers) are slightly optimistic [20].

7.6 Ethical and clinical considerations

NITSCI is a research prototype, not a medical device. Both datasets are public and de-identified. Several issues must be addressed before any clinical use. First, models trained on data from one population can perform worse, or fail silently, on others; algorithmic bias in health care has had measurable consequences [80], so subgroup performance by age, sex, ethnicity and camera should be reported. Second, the operating point should be set together with clinicians to reach a target sensitivity rather than to maximise accuracy, since a false negative delays treatment while a false positive costs a specialist visit. Third, probabilities from deep networks are often miscalibrated [62]; calibration should be checked before a probability threshold is used for referral. Finally, prospective external validation reported according to TRIPOD+AI [63] is required, as autonomous DR systems in clinical use have shown [10].

8 Conclusion and future work

8.1 Conclusion

This paper presented NITSCI, a compact network for five-level diabetic retinopathy grading from colour fundus photographs, designed for the common situation in which only a few hundred labelled images are available. The model places a cross-expert head of 1.40 M parameters on an EfficientNetV2-S backbone. A global expert relates all retinal regions through self-attention; a lesion expert receives the same features re-weighted by a lesion-attention map that is learned from grade labels alone; the two experts exchange information through bidirectional cross-attention; and an image-adaptive gate decides how much each contributes. A classification head and an ordinal regression head are combined into one continuous grade score, so that the decision respects the order of the ICDR scale.

The second contribution is the evaluation protocol. The model is pretrained on APTOS-2019, fine-tuned on IDRiD with stratified five-fold cross-validation, and applied as a twenty-member ensemble of fold models and flips. Every data-dependent decision, including the choice of decision rule, the four grade cut-points and the referral threshold, is made on out-of-fold predictions. The held-out test set is used once, and every result is reported with a bootstrap confidence interval; every ablation is compared with the full model by a paired exact test. This protocol avoids the optimistic bias that arises when thresholds or models are selected on the test set, a frequent problem in small medical datasets.

Under this protocol NITSCI achieved a five-class accuracy of **[acc5]**% (95% CI **[acc5ci]**) with a quadratic weighted kappa of **[qwk5]**, and a referable-DR AUC of **[refauc]** with sensitivity **[refsens]**% and specificity **[refspec]**% on the IDRiD test set. *[AUTHOR: One or two sentences on the most important ablation findings, for example the effect of APTOS pretraining and of the cross-expert head relative to the plain baseline, stated with their confidence intervals or p-values.]* These results indicate that careful use of a larger in-domain dataset, an architecture that focuses on lesion evidence without needing lesion annotations, and an ordinal decision rule can make DR grading on small datasets more reliable. They do not establish clinical readiness, which requires external and prospective validation.

Table 9 links each challenge identified in the Introduction to the component of NITSCI that addresses it and to the experiment that measures its effect.

8.2 Future work

Five directions follow directly from the limitations identified in Section 7:

1. **Official partition and benchmark comparison.** Re-running the pipeline on the official IDRiD grading partition will allow direct comparison with the challenge results [13].
2. **External validation.** Evaluating the trained ensemble without any re-training on independent datasets such as Messidor [35] and DDR [36], with subgroup analysis by camera and image quality, will measure how well the model generalises.
3. **Validated lesion attention.** The IDRiD lesion masks make it possible to test whether the learned lesion-attention map actually highlights lesions, for example with pointing-game accuracy, and to add weak lesion supervision when masks are available [39].
4. **Stronger initialisation.** Replacing the ImageNet-21k backbone with a retinal foundation model [34] may improve data efficiency further, and the cross-expert head can be attached to any backbone that produces a spatial feature map.
5. **Calibrated, uncertainty-aware referral.** Calibrating the referral probability [62] and using the disagreement between ensemble members as an uncertainty signal to defer difficult cases to a human grader [37, 58] would make the system safer to deploy in a screening workflow.

Table 9: Summary of the challenges addressed, the corresponding design choices, and where their effect is measured.

Challenge	Design choice in NITSCI	Evidence
Small, sparse lesion evidence	Lesion-attention map defining a lesion expert; bidirectional cross-attention with a global expert; adaptive gate	Ablations w/o lesion map, w/o cross-attention, w/o gate; plain baseline (Table 7)
Ordered labels	Classification + ordinal regression heads fused into one score; cut-points fitted on OOF data	Ablation w/o ordinal head; decision-rule comparison (Table 8); QWK, within-one-grade accuracy
Few labelled images	APTOS-2019 pretraining; EMA weights; five-fold ensemble with TTA	Ablation w/o APTOS; single model vs. ensemble (Table 8)
Camera and illumination differences	Ben Graham local-contrast normalisation	Ablation CLAHE (Table 7)
Class imbalance	Inverse-square-root class weights, label smoothing; balanced metrics	Balanced accuracy, macro F1, per-grade results (Table 5)
Optimistic evaluation	Test set used once; thresholds on OOF data; bootstrap CIs; McNemar tests	Tables 4–7

8.3 Recommendations for small-dataset grading studies

Beyond the specific model, our experience suggests four practices for studies that, like this one, work with a few hundred labelled images:

- **Lock the test set first.** Split it off before any experiment and record its indices; tune nothing on it, including thresholds and the choice between decision rules.
- **Report uncertainty, not only point estimates.** With fewer than one hundred test images, a bootstrap interval is often wider than the differences between methods; reporting it prevents over-claiming.
- **Compare under identical conditions.** Baselines and ablations should share splits, pre-processing, augmentation and inference; numbers copied from papers that used other splits are not comparable.
- **Release the pipeline.** Code that regenerates every number, including the test indices and per-image predictions, lets reviewers and other groups verify and extend the results.

Declarations

CRedit authorship contribution statement. *[AUTHOR: e.g. Name: Conceptualization, Methodology, Software, Investigation, Writing – original draft. Name: Supervision, Writing – review & editing.]*

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding. *[AUTHOR: Funding source and grant number, or “This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.”]*

Data availability. IDRiD and APTOS-2019 are publicly available from their providers [12, 24]. The test-set indices, per-image predictions and all metrics are produced by the released code.

Code availability. *[AUTHOR: Repository URL (e.g. GitHub + Zenodo DOI) containing the notebook, export cell and trained weights.]*

Ethics statement. This study used only publicly available, de-identified datasets; no new data were collected from human participants.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the authors used an AI assistant (Claude, Anthropic) to help draft and structure the manuscript text from the authors’ code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. *[AUTHOR: Keep this statement accurate; most publishers, including Elsevier, require it.]*

Acknowledgements. *[AUTHOR: Optional.]*

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Figure file `figures/distribution.pdf` not found.
Grade histograms of IDRiD and APTOS-2019 (export cell).

Figure S1: Grade distributions of IDRiD (target) and APTOS-2019 (pretraining).

Figure file `figures/curves.pdf` not found.
Per-fold training loss, validation accuracy and QWK.

Figure S2: Training loss, validation accuracy and validation QWK per fold during IDRiD fine-tuning.